

Counterfactual Event Bottlenecks for Leakage-Safe Financial Disclosure Modeling

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Abstract

Financial event models often collapse two distinct problems into one representation: ordinary market dynamics and the incremental impact of a newly disclosed event. We propose the Counterfactual Event Bottleneck Transformer (CEBT), an architecture that predicts a no-event baseline from pre-event market history and routes disclosure information through a compact stochastic bottleneck that predicts an event-induced residual. The model is trained with matched same-ticker no-event controls, a residual-suppression loss for controls, and strict temporal feature gates. On a real public-data experiment covering 7,236 SEC 8-K events from 98 usable large U.S. firms between 2020 and 2025, full CEBT does not dominate every baseline on raw regression error. Its evidence is narrower but more interesting: CEBT is the only tested model with positive held-out abnormal-return rank IC under iid, ticker-clustered, and month-clustered bootstrap intervals. In paired rank-IC comparisons, CEBT significantly improves over no-event dynamics and concat fusion, while its improvement over the no-control CEBT ablation is positive but not statistically resolved. CEBT also suppresses residual magnitude assigned to no-event controls. The main claim is not that CEBT is a deployable trading system, but that an explicit event bottleneck and no-event control objective produce a more temporally disciplined event-residual representation.

1 Introduction

Disclosure events are difficult modeling targets because market movement after a filing is not caused only by the filing. A stock’s subsequent return reflects market drift, sector movement, liquidity, firm-specific momentum, and the information content of the disclosure itself. Standard fusion models mix these sources into one hidden representation, making it hard to tell whether the model learned an event effect or simply reproduced ordinary dynamics.

We introduce CEBT, an event modeling architecture that decomposes the forecast into

$$\hat{y}_{t+h} = f_{\text{no event}}(x_{<t}) + g_{\text{event}}(z_e, x_{<t}),$$

where $f_{\text{no event}}$ is learned from pre-event numeric history and z_e is a compact stochastic disclosure bottleneck. The bottleneck is trained with matched no-event controls: dates for the same ticker that are far from known SEC 8-K events. For controls, the model is penalized when it assigns a large event residual. This makes the residual interpretable as a candidate event-specific increment rather than an unconstrained shortcut.

Our contributions are:

1. A no-event-plus-residual neural architecture for financial disclosure modeling.
2. A stochastic event bottleneck trained with matched no-event controls.
3. A leakage-safe public-data implementation using SEC accepted timestamps and public price histories.
4. A real-data study showing that residual control changes model behavior and that CEBT produces the strongest abnormal-return rank ordering under clustered uncertainty checks.

2 Related Work

CEBT is closest to causal and event-driven financial forecasting. Classical event-study methods estimate abnormal returns around public information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]; CEBT keeps this event-study grounding but learns a neural decomposition of baseline dynamics and disclosure residuals. CAMEF studies causal-augmented multi-modal financial forecasting [Chen et al., 2025]. Causal Transformer and G-Transformer estimate counterfactual outcomes in temporal settings [Melnychuk et al., 2022, Feng et al., 2024]. The architectural ingredients also connect to Transformers [Vaswani et al., 2017], variational latent-variable models [Kingma and Welling, 2013], and the information bottleneck principle [Tishby et al., 2000]. Financial-text work shows that disclosure language contains useful signal but is easy to confound with disclosure timing, firm identity, and market state [Loughran and McDonald, 2011, Cohen et al., 2020, Araci, 2019]. Time-series foundation models such as Chronos, Time-LLM, MOMENT, and Mamba improve sequence modeling capacity but do not isolate disclosure-induced residuals [Ansari et al., 2024, Jin et al., 2023, Wu et al., 2024, Gu and Dao, 2023]. Financial LLM benchmarks and agents test retrieval or question-answering over filings rather than a new event-residual architecture [Islam et al., 2023, Jiang et al., 2026]. Leakage-focused work such as Profit Mirage motivates strict temporal evaluation in financial agents [Li et al., 2025]. Subliminal-learning results further motivate caution around hidden outcome signal in generated or compressed representations [Cloud et al., 2025].

3 Method

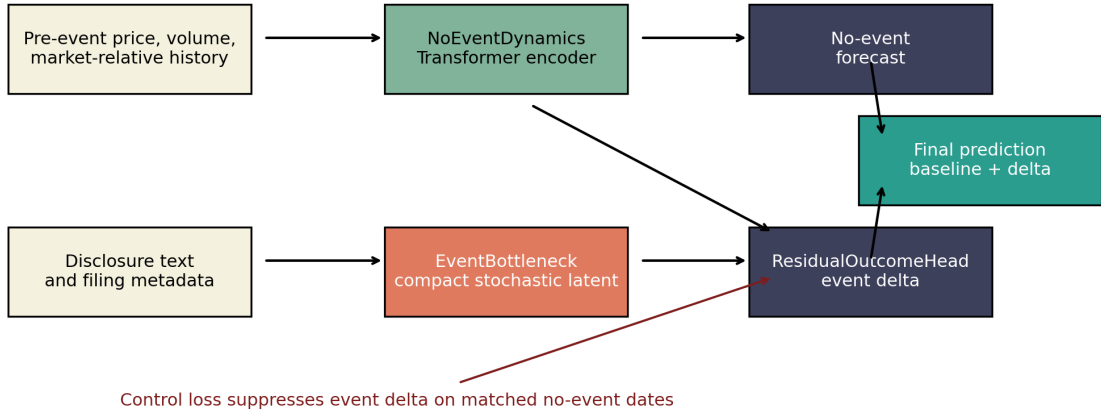


Figure 1: CEBT decomposes a forecast into no-event dynamics and an event-induced residual routed through a compact stochastic bottleneck. Matched no-event controls penalize residuals on dates where no disclosure event should be needed.

Notation and estimand. Each sample i has a label start date τ_i , a 40-trading-day pre-event tensor $X_i \in \mathbb{R}^{40 \times 8}$, an event embedding $e_i \in \mathbb{R}^{256}$, metadata $m_i \in \mathbb{R}^6$, a control indicator c_i , and a target vector $y_i \in \mathbb{R}^3$. The three targets are five-trading-day market-adjusted return, volatility jump, and volume jump. CEBT estimates a predictive event residual, not an identified structural treatment effect. The counterfactual claim is therefore modest: matched no-event rows define a placebo pressure that encourages the residual channel to be small when no disclosure is present, while real causal identification would require stronger assumptions and intraday or quasi-experimental evidence.

NoEventDynamics. The no-event module receives a fixed pre-event window of price, volume, market, and relative-return features. A Transformer encoder pools this history and predicts the ordinary forward movement expected without relying on the event text.

EventBottleneck. The event module embeds the disclosure text and metadata, then parameterizes a Gaussian latent $q_\phi(z_i | e_i, m_i) = \mathcal{N}(\mu_i, \text{diag}(\exp(\ell_i)))$. During training, $z_i = \mu_i + \epsilon \exp(\ell_i/2)$, $\epsilon \sim \mathcal{N}(0, I)$; at evaluation, we use μ_i . The prior is $p(z) = \mathcal{N}(0, I)$. The latent dimension is intentionally small: the paper-scale model uses eight latent coordinates, hidden width 96, two Transformer encoder layers, four attention heads, and dropout 0.10. We add a KL term and sparsity term so the event representation cannot behave like an unrestricted text embedding.

ResidualOutcomeHead. The residual head predicts an event-induced delta from the no-event state, metadata, and z_i . The final forecast is the sum of the no-event forecast and this residual:

$$\hat{y}_i = b_\theta(X_i, m_i) + \Delta_\theta(s_i, z_i, m_i), \quad (1)$$

where s_i is the pooled no-event state.

Training objective. The loss combines supervised MSE, a no-event control residual penalty, KL regularization, latent sparsity, and a small residual-variance consistency penalty:

$$\mathcal{L} = \lambda_y \|\hat{y} - y\|_2^2 + \lambda_c \mathbb{K}_{control} \|\Delta_e\|_2^2 + \lambda_{kl} D_{KL}(q(z_e) \| p(z)) + \lambda_s \|z_e\|_1 + \lambda_v \text{Var}(\Delta_e). \quad (2)$$

For the paper-scale run, $\lambda_y = 1.0$, $\lambda_c = 0.25$, $\lambda_{kl} = 0.01$, $\lambda_s = 0.001$, and $\lambda_v = 0.05$. The control penalty is averaged over no-event rows in each batch. The consistency term is the batch variance of the residual output and is included only as a small stabilizer, not as the main source of the result.

4 Data and Leakage Controls

We build a public-data experiment from SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The paper-scale run covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and daily public prices. Tickers are mapped to CIKs through the SEC company-tickers file. Filings are read from SEC submissions JSON, including archived submissions, using the SEC `acceptanceDateTime` field and the primary document URL in the EDGAR archive. The text parser removes scripts, styles, and markup with BeautifulSoup and stores the extracted text plus SHA-256 provenance. Each event row stores CIK, ticker, accession number, form type, filing date, accepted timestamp, available timestamp, source URL, text hash, and local provenance.

Daily prices are downloaded from Stooq first and Yahoo’s public chart endpoint as a fallback. SPY is the market proxy for abnormal returns. The trading calendar is inferred from observed price tables rather than from an invented holiday file. Event label windows start from the first tradable session after public availability. If a filing is accepted after 16:00 ET, or on a non-trading day, the label starts on the next observed trading day.

Features use only prices strictly before the event label start date. The eight numeric features are one-day return, five-day compounded return, 20-day compounded return, 20-day realized volatility, 20-day volume z-score, market return, market-relative return, and close-to-open return. Labels use future returns only as targets. We construct one deterministic same-ticker no-event control for each filing, excluding dates within a 10-calendar-day blackout window around known events. Control rows receive a zero event embedding and explicit control metadata; this makes the control objective clean for a first test, but it is also a limitation discussed below.

Table 1: Paper-scale public-data substrate.

Artifact	Count
Usable companies	98
SEC 8-K events	7,236
Matched no-event controls	7,236
Feature rows	14,472
Public price rows	159,093
Held-out test rows	2,463

5 Experiments

The paper-scale run uses `configs/paper_v3.yaml` with seed 29. Rows with $\tau_i \leq 2023-12-31$ are training rows, rows with $\tau_i \leq 2024-12-31$ are validation rows, and later rows are held out. This gives 9,729 training rows, 2,280 validation rows, and 2,463 held-out rows across 98 held-out tickers and 13 held-out label-start months. All models are trained for 15 epochs with AdamW, learning rate 10^{-3} , weight decay 10^{-4} , batch size 64, and mixed precision on CUDA when available. No early stopping or hyperparameter search is used for the reported run.

We compare five models: no-event dynamics only, text-only MLP, concat fusion, CEBT without control losses, and full CEBT. The no-event baseline sees X_i and metadata but no disclosure embedding. The text-only baseline sees e_i and metadata but no price history. Concat fusion uses the same no-event encoder as CEBT, then directly concatenates the pooled state, event embedding, and metadata without a stochastic bottleneck or additive residual constraint. CEBT without controls has the same architecture as full CEBT but sets $\lambda_c = 0$.

Rank IC is Spearman correlation between predicted and realized abnormal return over held-out rows. The main rank-IC value is computed across all held-out rows; Figure 4 additionally shows monthly event-only rank IC buckets with at least eight disclosures. Confidence intervals use 1,000 percentile bootstrap resamples. Row bootstraps sample held-out rows with replacement. Ticker-clustered and month-clustered intervals sample clusters with replacement and concatenate all rows in sampled clusters. Paired intervals resample aligned prediction rows and report the difference in the relevant metric.

Table 2: Held-out paper-scale results. MSE is over the three target dimensions. Rank-IC intervals shown here are ticker-clustered; month-clustered intervals are discussed in the text.

Model	MSE	95% CI	Rank IC	Ticker-cluster 95% CI
No-event	0.0696	[0.0535, 0.0907]	-0.0411	[-0.0840, 0.0016]
Text-only	0.0643	[0.0512, 0.0820]	-0.0009	[-0.0456, 0.0434]
Concat fusion	0.0597	[0.0456, 0.0784]	-0.0182	[-0.0597, 0.0267]
CEBT, no controls	0.0666	[0.0507, 0.0874]	0.0279	[-0.0139, 0.0716]
CEBT	0.0660	[0.0504, 0.0867]	0.0463	[0.0052, 0.0893]

Table 3: Paired MSE comparisons against full CEBT. Positive values mean CEBT has lower MSE than the baseline.

Baseline	Baseline-CEBT MSE	95% CI	CEBT better?
CEBT, no controls	0.00056	[0.00019, 0.00097]	Yes
Concat fusion	-0.00630	[-0.00863, -0.00416]	No
No-event	0.00359	[0.00184, 0.00549]	Yes
Text-only	-0.00171	[-0.00524, 0.00143]	No

Table 4: Paired abnormal-return rank-IC comparisons against full CEBT. Positive values mean CEBT has higher rank IC than the baseline.

Baseline	CEBT-baseline rank IC	95% CI	CEBT better?
CEBT, no controls	0.0184	[-0.0202, 0.0595]	No
Concat fusion	0.0645	[0.0058, 0.1215]	Yes
No-event	0.0874	[0.0251, 0.1534]	Yes
Text-only	0.0472	[-0.0097, 0.1028]	No

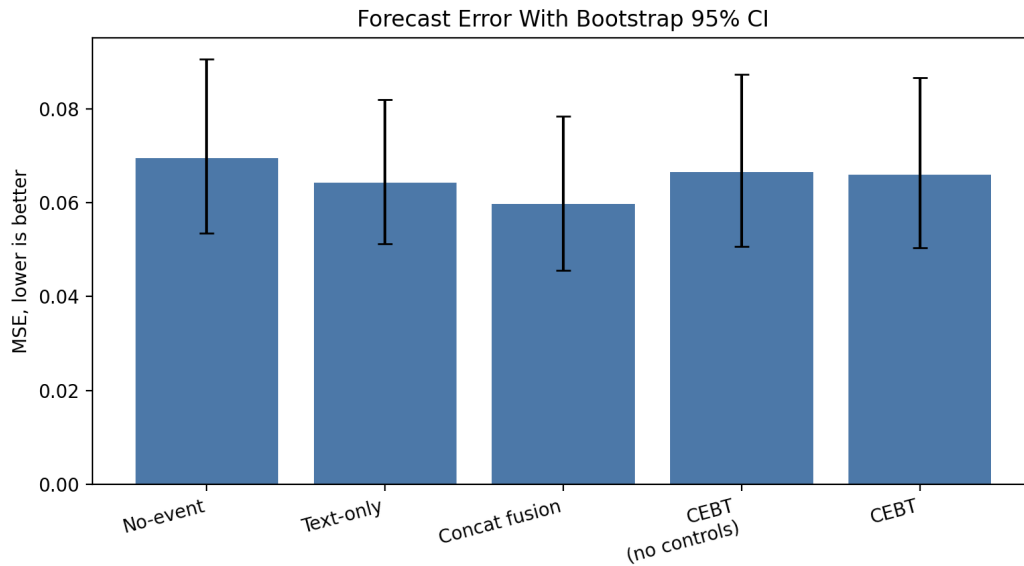


Figure 2: Forecast error with bootstrap confidence intervals. Concat fusion has the strongest point-estimate MSE; CEBT's advantage appears in rank ordering and residual discipline rather than raw multi-target regression error.

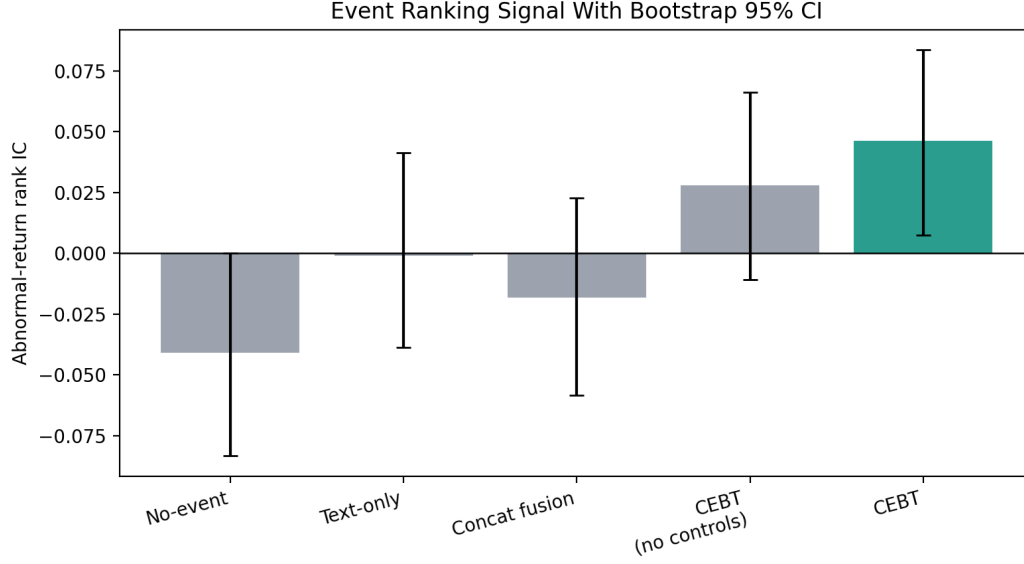


Figure 3: Abnormal-return rank IC with bootstrap confidence intervals. Full CEBT is the only tested model whose interval is entirely above zero.

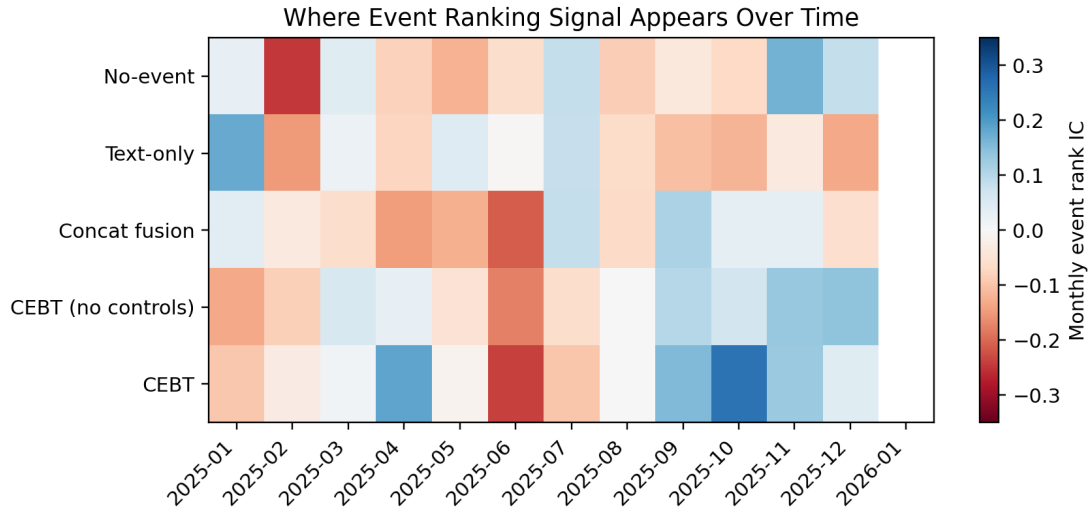


Figure 4: Monthly event-ranking signal by model. The heatmap shows where rank signal appears and whether it is temporally concentrated or persistent across the held-out period.

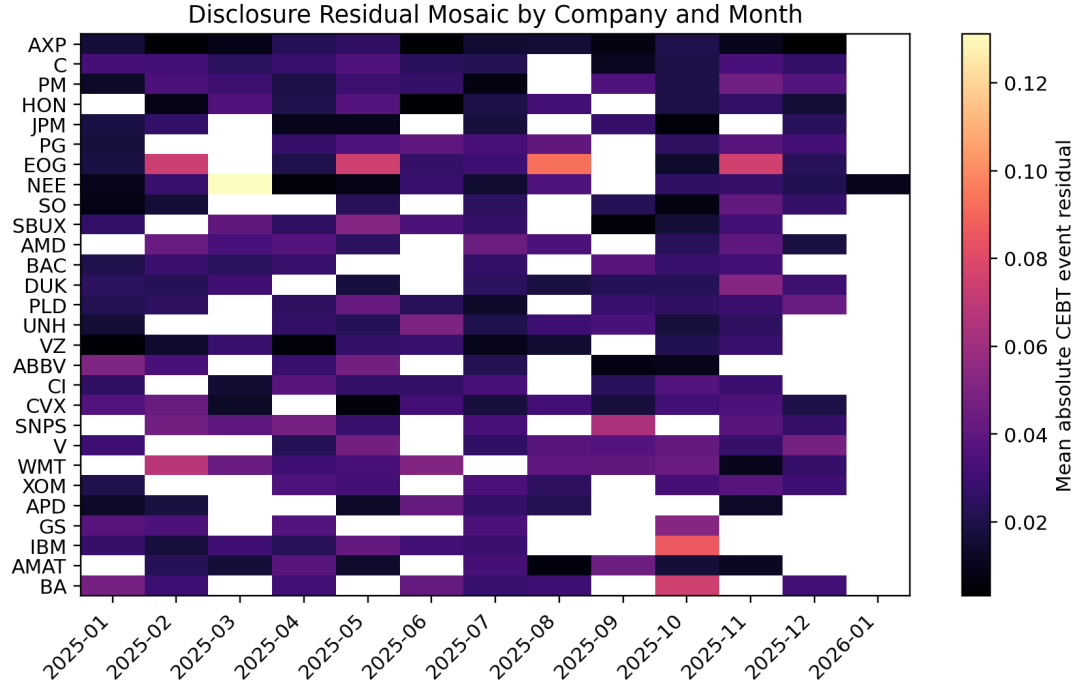


Figure 5: CEBT event-residual mosaic by company and month. Darker cells indicate months where the model routes more mass through the event residual rather than the no-event baseline; white cells indicate no held-out real 8-K events for that company-month.

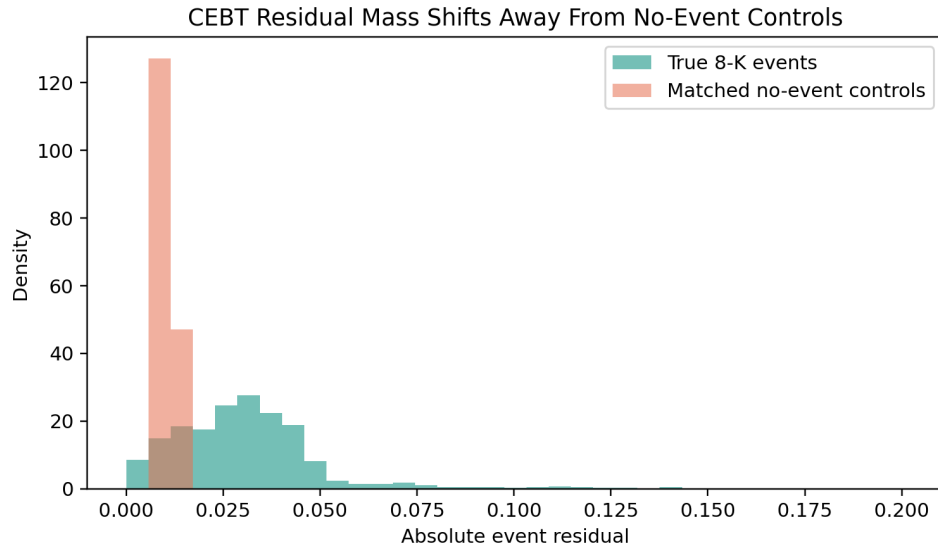


Figure 6: Residual distribution for true 8-K events and matched no-event controls. The distributional view is more informative than a mean-only bar chart: controls concentrate near zero, while true events have a heavier residual tail.

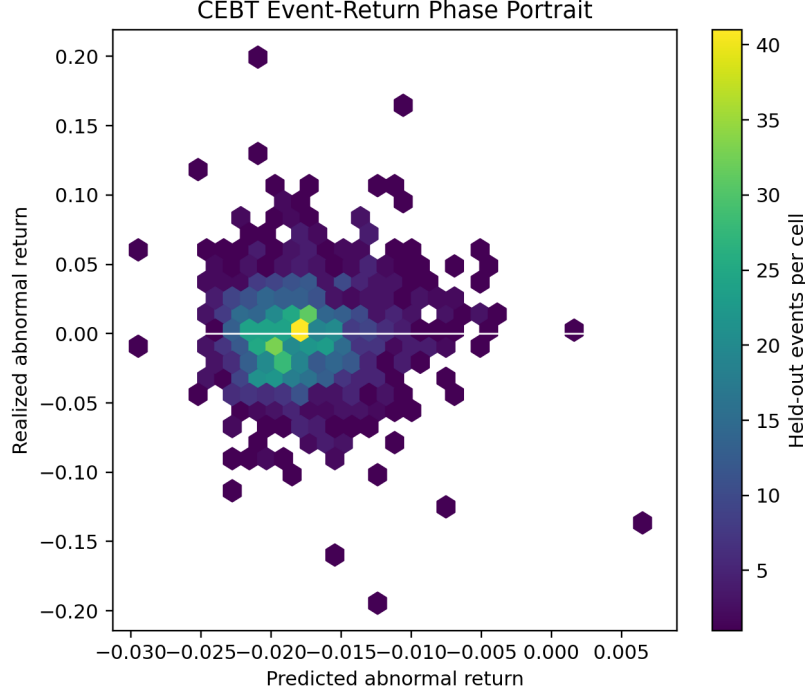


Figure 7: CEBT event-return phase portrait on held-out real disclosures. The density plot shows where predictions and realized abnormal returns concentrate, making the narrow prediction band and wide realized-return dispersion visible without plotting thousands of overlapping points.

6 Findings

CEBT is an event-ranking model, not the best raw regressor. As shown in Table 2 and Figure 2, concat fusion has the best point-estimate MSE, 0.0597, while full CEBT has MSE 0.0660. This matters: the architectural decomposition does not automatically minimize multi-target regression error. The evidence for CEBT is instead concentrated in rank ordering of event outcomes.

The event-ranking signal survives clustered uncertainty checks. Full CEBT has abnormal-return rank IC 0.0463 in Table 2. Its iid bootstrap interval is $[0.0074, 0.0836]$, the ticker-clustered interval is $[0.0052, 0.0893]$, and the month-clustered interval is $[0.0112, 0.0768]$. No other tested model has a positive clustered interval excluding zero. Figure 4 shows that this signal is not a single company-month artifact, although it is still modest in magnitude.

Paired rank comparisons separate CEBT from stronger baselines. CEBT improves paired rank IC over concat fusion by 0.0645 with a 95% paired interval $[0.0058, 0.1215]$, and over no-event dynamics by 0.0874 with interval $[0.0251, 0.1534]$. The improvement over CEBT without controls is positive, 0.0184, but not resolved by the paired interval $[-0.0202, 0.0595]$.

Control losses alter the residual semantics. The full CEBT model assigns mean absolute event residual 0.0313 to true events and 0.0103 to controls. The no-control variant assigns 0.0396 to true events and 0.0298 to controls. Thus, the control objective reduces spurious residual assignment to no-event dates by about 66%, while preserving a roughly 3.0x event-to-control residual ratio. Figures 5 and 6 show the same behavior visually: residual mass is sparse across company-months and shifted away from placebo rows.

7 Reproducibility

The reported run was produced with `configs/paper_v3.yaml` and `configs/model/cebt_small.yaml`. The pipeline order is:

`10_build_events` `20_download_prices` `30_build_features` `60_ablate` `70_make_tables`

The H200 run used PyTorch 2.11.0 with CUDA 12.8. The code stores raw SEC metadata, text hashes, price rows, feature metadata, model checkpoints, predictions, table CSVs, and generated figures. The automated tests check timestamp gating, after-hours label starts, feature leakage rejection, control construction, deterministic cached tensors, finite forward losses, and evaluation leakage guards. No paid data, closed-source model endpoint, or generated label is used for the main result.

8 Limitations

This is a first paper-scale run, not a final investment model. The text encoder is a deterministic open hashing baseline rather than a large pretrained financial encoder. Control rows use zero event embeddings and explicit control metadata, so residual suppression on controls is not yet a hard placebo-text test. The public price source is not CRSP, and labels use daily closes rather than intraday event windows. The universe is large-cap biased and may contain survivorship effects, ticker-history issues, and 8-K item-mix sensitivity. The experiment uses SPY-adjusted returns rather than richer factor-model residuals and excludes transaction costs. Two configured tickers did not yield final usable event rows because of SEC retrieval failures during the build. The counterfactual language should be read as architecture and regularization, not full causal identification. The next version should test pretrained open embeddings, 10-Q/10-K extensions, sector and factor controls, and harder placebo controls where no-event dates receive unrelated historical disclosure text instead of a zero event embedding.

9 Ethics and Use

This paper studies event representations using public filings and public daily prices. It is not a trading recommendation system, and the reported rank signal is too small and too coarsely measured to justify deployment. The release should include code, configs, and provenance metadata so that future users can audit leakage and reproduce or falsify the claims.

10 Conclusion

CEBT reframes event-driven financial modeling as a decomposition problem: ordinary market dynamics plus a disclosure-induced residual. In a real public-data experiment, CEBT does not beat concat fusion on raw MSE, but it produces the strongest abnormal-return rank ordering and remains positive under ticker- and month-clustered uncertainty checks. The result supports CEBT as a promising architecture for studying event-specific effects under strict temporal information boundaries.

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